

Emergent

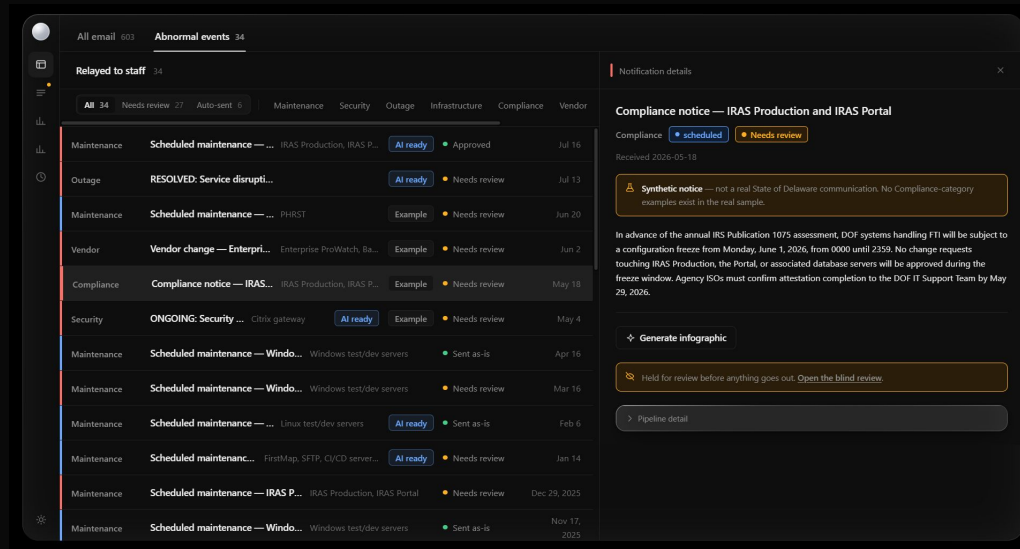
Email delivery **reinvented.**



Too many critical and abnormal notifications. Too little attention.

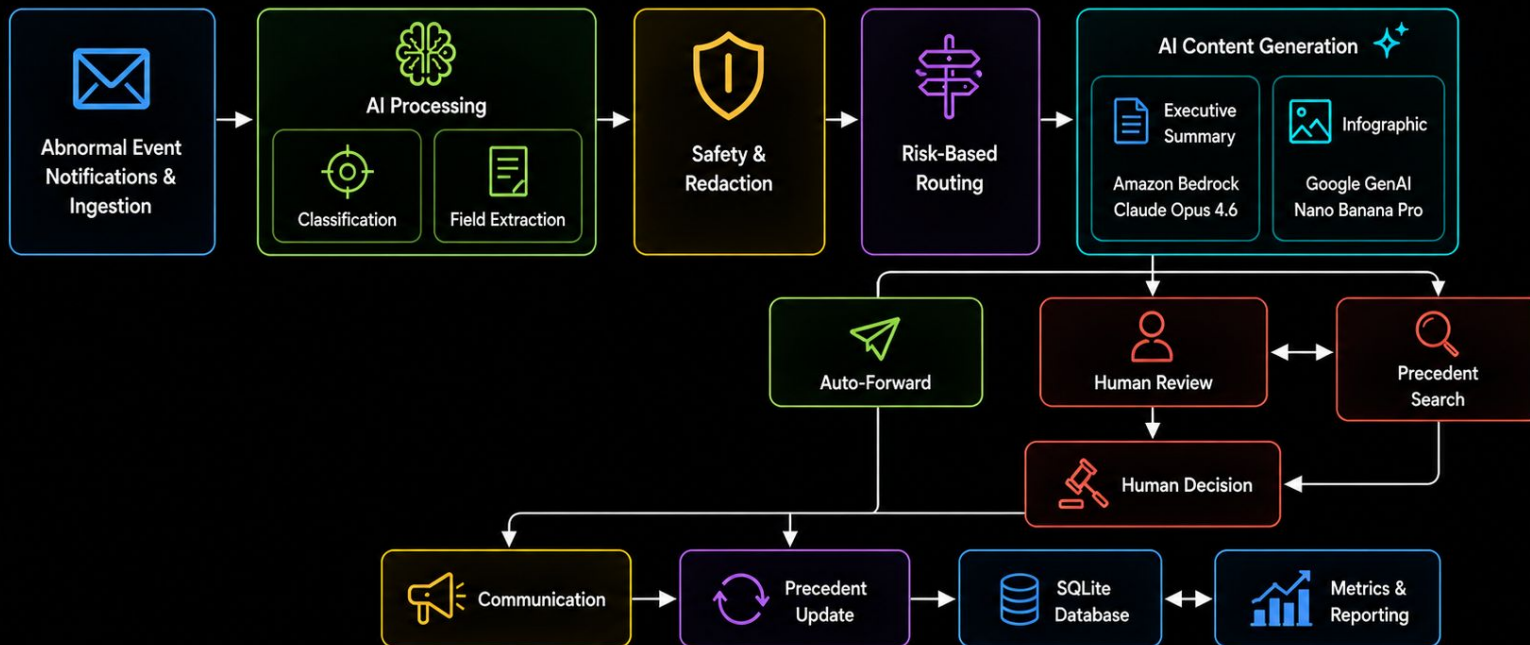
- **600 notifications in just 2 weeks**
Employees get buried in abnormal-event emails on top of their normal workload.
- **Notification fatigue**
When alerts never stop, they blur together. Important ones get skimmed, ignored, or missed.
- **Small interruptions add up**
Even **1–2 minutes** per notification means roughly **10–20 hours** spent reading, interpreting, and figuring out what matters across 600 messages—instead of doing core work.
- **Missed alerts have consequences**
A missed notice means skipped required actions, systems left running through maintenance, or being caught off guard by an outage.

An AI pipeline that classifies incoming IT notices, drafts employee and leadership-facing comms, and routes everything through genuine human approval.



It matches how a reviewer already thinks: rules for what's certain, judgment for what isn't.

Emergent – AI-Powered Abnormal Event Notification System



Next.js



React



TypeScript



Tailwind CSS



SQLite



Amazon Bedrock
Claude Opus 4.6



Google GenAI
Nano Banana Pro

Responsible AI & Data

DATA USED

- 603 real subject lines (1 week, tagged)
- 226 full real notices (19 months)

KNOWN LIMITATIONS

- Every accuracy number here is model-generated labels agreeing with model-generated answers
- Not all real data has been blind-labeled by humans
- A true accuracy figure compared to human labels comes with usage
- Tradeoff: Higher-quality visual generation can require more capable models, more iteration, and greater compute/cost, making fully generated text-heavy graphics challenging

SAFEGUARDS BUILT BEYOND AN API CALL

- For High risk messages, AI-generated content is never auto-sent
- No prompt injection→A trained classifier with real weights, not a prompted LLM
- Enforced in the routing logic and again at the approval endpoint. not as a setting anyone can turn off

SYNTHETIC COVERAGE

[illegible]

 Abnormal Events_004.xlsx Excel Spreadsheet	
Date	Notes
01-01-20	AMS production and the Partial will be undergoing new changes from Monday, January 13th, starting from 6:00 AM to 10:00 AM. All new orders should be made a minimum of 10 business days in advance.
01-03-20	On Sunday, January 26, 2020, from 11:00 AM to 1:00 PM, there will be performance maintenance on the AMS production Line series which will impact the available inventory applications.
01-03-20	On Monday, January 27, 2020, from 12:00 PM to 1:00 PM, there will be performance maintenance on available orders. To avoid a partial event, contact us at info@wms.com .
01-13-20	On Monday, January 27, 2020, from 10:00 AM to 12:00 PM, there will be working maintenance on the AMS production Line series which will impact the available inventory applications.
01-14-20	AMS production and the Partial will be undergoing new changes from Monday, January 28, starting from 6:00 AM to 10:00 AM. All new orders should be made a minimum of 10 business days in advance.
01-14-20	On Monday, January 28, 2020, from 12:00 PM to 1:00 PM, there will be performance maintenance on available orders. To avoid a partial event, contact us at info@wms.com .
01-14-20	The maintenance activities for the AMS will be new released and new.
01-14-20	On Monday, January 28, 2020, from 12:00 PM to 1:00 PM, there will be performance maintenance on available orders. To avoid a partial event, contact us at info@wms.com .
01-16-20	On Monday, January 28, 2020, from 10:00 AM to 12:00 PM, there will be working maintenance on the AMS production Line series which will impact the available inventory applications.
01-17-20	On Monday, January 27, 2020, from 11:00 AM to 1:00 PM, there will be performance maintenance on the AMS production Line series which will impact the available inventory applications.
01-18-20	AMS production and the Partial will be undergoing new changes on Monday, January 29, 2020, starting from 6:00 AM to 10:00 AM. All new orders should be made a minimum of 10 business days in advance.
01-19-20	AMS production and the Partial will be undergoing new changes from Monday, January 30, starting from 6:00 AM to 10:00 AM. All new orders should be made a minimum of 10 business days in advance.
01-20-20	On Monday, January 27, 2020, from 11:00 AM to 1:00 PM, there will be performance maintenance on the AMS production Line series which will impact the available inventory applications.

Raw data is messy and unevenly biased. To level unrepresentative bias and obtain enough data for accurate fine-tuning, we created synthetic data for validation and training.

FOR PRODUCTION

- A human-validated labeling pass
- Scoped mailbox integration
- A named approval owner
- Monitoring for reviewer disagreement over time

Compute and Providers



SERVICES USED

AWS Bedrock, Kiro, Vercel, Claude, Github Repository, and Nano Banana through Google Cloud

WHAT RAN THERE

API integration, Application Hosting & Demo Support, Code Refinement, Infographic Design, and GitHub Code Management

WORKED / DIDN'T

- Kiro ↔ Google Cloud setup was difficult: repeated authorization, command, and VM connection failures
- Image-generation limitations: Bedrock image outputs sometimes produced blurry text, misspellings, altered wording, and inconsistent layout

WHAT WE LEARNED

A lot of services have already been thought out and engineered, so instead of reinventing the wheel, it's better to build off of the design and work with all the wonderful providers we have the honor to be with.

WHY EMERGENT?

- Greater awareness of critical events
- Faster understanding, fewer missed details
- Human accountability stays in the loop
- Less disruption to employees' core work

WHO BENEFITS?

- Employees and IT / DTI teams

TIME • COST • QUALITY

- Less drafting, summarizing & formatting
- Staff effort shifts from repetitive communication work to higher-value tasks
- More consistent messages, clearer actions & fewer missed details

PATH TO PILOT

- Start with one agency, Use approved templates + historical notifications, Assign a human reviewer / approver, Measure time saved, accuracy & communication quality

- **PROBLEM**

- Too much information, too little attention

- **SOLUTION**

- AI turns noise into what matters
Classify, extract, summarize, personalize, and keep humans in control.

- **RESULT**

- Clearer communication, faster understanding
Important information becomes easier to find, understand, and act on.

- **IMPACT**

- Less time managing information. More time doing meaningful work.
For Delaware, that means more productive employees and fewer missed critical alerts.

- **ONE STEP FURTHER**

- We started by helping Delaware employees miss fewer critical alerts. The bigger opportunity is helping everyone spend less time managing information, and more time on what matters.
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